

ANTHROPOGENIC IMPACTS AND THEIR INFLUENCE ON THE SPATIAL DISTRIBUTION OF THE ODONATA OF WADI EL HARRACH (NORTH-CENTRAL ALGERIA)

Mouna HAFIANE¹, Djamila HAMZAOU¹, Fouzia ATTOU¹, Djaouida BOUCHELOUCHE¹
Abdeslem ARAB¹, Ahmed H. ALFARHAN² & Boudjéma SAMRAOUI^{2,3}

¹ University of Sciences and Technology Houari Boumédiène. Faculty of Science. Biological, Dynamic and Biodiversity Laboratory, BP 32, El Alia, Algiers, Algeria. E-mails: mounahafiane@yahoo.fr, hamzaoui_djamila@hotmail.com, attou.fouzia@yahoo.fr, bouchelouche_djaouida@hotmail.com, abdeslama@yahoo.fr

² Center of Excellence for Research in Biodiversity, King Saud University, Saudi Arabia. E-mail: bsamraoui@yahoo.fr

³ Department of Biology, University of Annaba and Laboratoire de Conservation des Zones Humides, University of Guelma, Algeria.

RESUME.— *Impacts des facteurs anthropiques et leur influence sur la distribution spatiale des Odonates de l'oued El Harrach (Centre-Nord de l'Algérie).*— Malgré une connaissance relativement bonne de l'odonatofaune de l'Algérie et du Maghreb, de vastes régions comme le centre de l'Algérie demeurent peu explorées. Ces régions ont connu une explosion démographique et leur développement agro-industriel, au cours du siècle passé, a impacté défavorablement la plupart des écosystèmes naturels. L'oued El Harrach, un cours d'eau saisonnier qui traverse la plaine de la Mitidja, près d'Alger, est notoirement connu pour sa pollution et il représente ainsi un cas intéressant pour évaluer les impacts anthropiques sur les peuplements de macro-invertébrés dulçaquicoles. L'utilisation de plus en plus fréquente des méthodes d'évaluation mises au point pour le suivi et l'évaluation de l'intégrité écologique des cours d'eau, nous a encouragés à échantillonner les Odonates de l'oued El Harrach et mesurer divers paramètres physico-chimiques représentatifs de la qualité de l'eau du milieu. Les deux jeux de données ont été analysés conjointement pour établir une correspondance entre les deux évaluations. Les résultats indiquent une congruence entre les deux approches et soulignent le potentiel des Odonates comme bio-indicateurs utiles des conditions écologiques des écosystèmes lotiques.

SUMMARY.— In spite of a relatively good knowledge of the odonatofauna of the Maghreb, some large areas like the centre of Algeria have remained little explored. These areas have witnessed a demographic explosion and an agro-industrial development over the last century which have adversely impacted most natural ecosystems. Wadi El Harrach, an intermittent river that cuts through the Mitidja plain, near Algiers, is notorious for its pollution and represents a good model to investigate anthropogenic impacts on freshwater communities. In line with the increasing use of assessment methods developed to monitor and evaluate the ecological integrity of running waters, we sampled Odonata of Wadi El Harrach and measured various abiotic parameters representative of the water quality of the habitat. Both data sets were jointly used to analyse the correspondence between the two assessments. Results indicated congruence between both approaches and highlighted the potential of Odonata as reliable bioindicators of ecological conditions of lotic ecosystems.

In the light of global and local habitat alterations, there is a real need to assess the consequences of ecosystem change for human well-being and develop management strategies to conserve and enhance the sustainable use of ecosystems (Millennium Ecosystem Assessment, 2005). Freshwater ecosystems are under severe pressure worldwide (Abell, 2002) and, in North Africa, they will probably face increasing alterations, demands and depletions. Human encroachment on northern African wadis, as happened with rivers, worldwide (Dynesius & Nilsson, 1994), has been unsparing. Water extraction, pollution, dam construction, and channelization have profoundly modified the riverine landscape and altered its function.

One important step in the management of ecosystems is a good knowledge of patterns in species diversity and their driving processes. Insects are a successful group of living organisms that are essential to the functioning of many terrestrial and freshwater ecosystems (Schminke, 2007). Odonata, perceived as charismatic insects, have been deemed to provide direct benefit to

humans (Simaika & Samways, 2008) through their contribution to ecosystem services to humans in four distinct categories: provisioning, cultural, supporting, and regulating services, recognized by the authoritative document known as the Millenium Ecosystem Assessment (2005).

Benthic macroinvertebrates have been widely used for monitoring aquatic habitats, but Odonata, both as adults and larvae, have proven excellent surrogates and may outperform macroinvertebrates as bioindicators of freshwater health and ecological integrity (Oertli, 2008; Monteiro-Junior *et al.*, 2015; Willigalla & Fartmann, 2012) and by providing reliable assessments of the hydrology and habitat alterations of fluvial systems (Ferrerias-Romero *et al.*, 2009).

Despite its geographical position near Algiers, and notoriety as one of the most polluted watercourse in North Africa, Wadi El Harrach has so far received little attention from researchers. Likewise, the benthic macrofauna, increasingly used to monitor freshwater health, of the Algerian watercourses is still relatively poorly known, but recent progress has been made (Thomas & Lounaci, 1989; Lounaci *et al.*, 2000a, b; Arab *et al.*, 2004; Chaib *et al.*, 2013a, b). Recent research has mainly focused on the larval stages (Cheraïria *et al.*, 2014) with few studies focused on the adult stages (Annani *et al.*, 2012). Odonatology is an exception with work spanning over a century and a half (Samraoui & Menai, 1999; Boudot *et al.*, 2009). In recent work, more emphasis was put on using lotic Odonata as indicators of ecosystem integrity (Bouchelouche *et al.*, 2015; Hamzaoui *et al.*, 2015).

Knowledge of the status, distribution and life history of riverine Odonata, acting as umbrella species (Betrus *et al.*, 2005), as well as the identification of threats to their habitats are urgently needed to set up conservation and restoration strategies in view of the fast degradation of watercourses within the Mediterranean landscape (Olson & Dinerstein, 1998). Thus, there is a real need to survey North African riparian biodiversity and monitor changes through the use of rapid, low-cost methods (Chovanec & Waringer, 2001; Simaika & Samways, 2009). The identification of the physical habitat factors that influence Odonata abundance and distribution can facilitate the development of such tools. In this context, we set up to survey the Odonata of Wadi El Harrach and attempt to identify environmental determinants of their distribution.

MATERIALS AND METHODS

STUDY AREA

Wadi (or Oued) El Harrach is located in the northern center of Algeria. It springs in the Blidian Atlas, crosses the plain of Mitidja, and reaches the bay of Algiers. The length of the study area spans 40 km from the village Magtaa Lazreg to Baba Ali. The local climate is Mediterranean with hot, dry summers and wet, mild winters.

Upstream of the confluence of Wadi Magtaa and Wadi Lakhra (Fig. 1), the region has been deserted over the last twenty years due to security reasons. The upper part reaches of the wadi, between the two villages of Magtaa and Hammam Melouane is often used as a holiday resort in summer and the area is managed (clearance of vegetation) to that effect. The middle course is mainly impacted by agriculture (use of pesticides) whereas the lower course is the receptacle of industrial effluents. In this part of the wadi, the bed is channelled and the banks are heavily planted with *Eucalyptus* trees.

SAMPLING AND METHODOLOGY

Monthly sampling was carried out from March 2013 to February 2014. Records of adult Odonata were noted at 8 stations (S1-S8) selected along the main watercourse, and distributed between altitudes 25-220 m (Fig. 1, Tab. I). The two upstream stations S1 and S2 are both located on the two tributaries whose confluence gives birth to Wadi El Harrach. The most notable difference between S1 and S2 is the size of the boulders (larger in S1) that litter the watercourse. The surrounding vegetation of both stations is dominated by plantations of *Pinus halepensis* and more often than not the native *Nerium oleander* gives way to the invasive *Arundo donax*. S3 and its surroundings are used as recreational areas by holidaymakers during the summer months. These parts of the wadi are managed as swimming pools. The water quality of all the stations located downstream of S3 decreases markedly due to the increase of human encroachment (presence of villages). At S4, the wadi is the receptacle of discharges from Hammam Melouane, a thermal resort. The middle and downstream parts of the wadi are characterized by the straightening of their banks and the loss of their native riverine vegetation replaced by *Eucalyptus* trees. The next two stations, S5 and S6, are sandwiched between Hammam Melouane and Bougara, where the Mitidja plain starts. This is the most fertile agricultural land in Algeria and intensive agriculture is

the hallmark of the surroundings of S7 and S8. At Baba Ali and neighbourhood, the wadi is the receptacle of residues of chemical fertilizers and pesticides that wash off from the adjacent fields. In these two stations, household waste becomes more frequent and conspicuous.

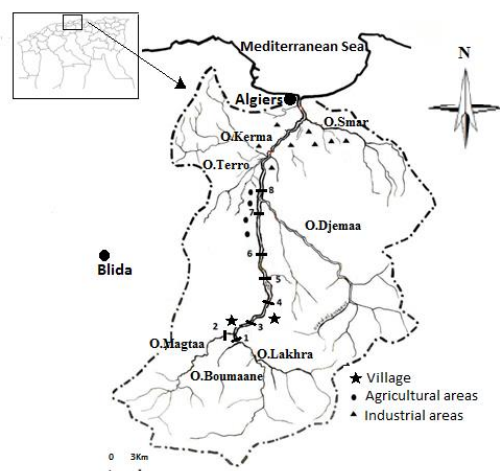


Figure 1.— Localization of sampled stations along Wadi El Harrach.

TABLE I
List of surveyed stations

Stations code	Stations (locality)	Altitude (m)	Latitude	Longitude
S1	MagtaaLazreg	220	36°27'53.30"N	3°0'54.82"E
S2	Lakhra and Boumaane	219	36°28'15.99"N	3°0'24.97"E
S3	Hammam Melouene 1	170	36°28'59.75"N	3°2'31.57"E
S4	Hammam Melouene 2	158	36°29'22.32"N	3°3'06.91"E
S5	Bougara 1	119	36°30'41.05"N	3°4'05.57"E
S6	Bougara 2	92	36°32'6.34"N	3°3'44.83"E
S7	Baba Ali 1	29	36°37'22.02"N	3°3'48.35"E
S8	Baba Ali 2	25	36°37'32.75"N	3°3'44.75"E

Sampling of Odonata was performed by walking slowly and repeatedly along a transect (100 m). Flying adults were often identified on sight, but some were caught using a butterfly net and these voucher specimens were carried to the laboratory to be identified (Dijkstra & Lewington, 2006; Grand & Boudot, 2007) and stored.

Field data collection was also carried out using a WTW 340i multiparameter to monitor physical and chemical variables: temperature (°C), pH, electric conductivity (µS/cm), salinity (‰), dissolved oxygen (mg/l). Other environmental parameters were also measured, such as current velocity (cm/s), bed's width, water depth, abundance of submerged and shore vegetation and shading/sun exposure.

STATISTICAL ANALYSIS

Analysis of the relationship between odonate species and habitat characteristics was carried out using an Ascending Hierarchical Classification (ACH) and a Canonical Correspondence Analysis (CCA). These analyses were made using the program R (R Development Core Team, 2014).

RESULTS

As might be expected, the results exhibited a strong upstream-downstream gradient of water characteristics. Dissolved oxygen exhibited a spatio-temporal variation with lower oxygen content in the downstream part (Tab. II), mainly during summer. The level of oxygen is known to decrease with rising temperature and dissolved mineral salts reduce the oxygen solubility. Very low values of dissolved oxygen (1.52 mg/l and 0.69 mg/l) were recorded during the summer at S7 and S8,

respectively. The high values of salinity and conductivity recorded at S4 are explained by the discharges of the thermal resort of Hammam Melouene which is located upstream of this station whereas those of S7 and S8 are due to agricultural and industrial effluents combined to increased domestic sewage (Tab. II).

TABLE II
Environmental variables measured at Wadi El Harrach

Stations		T° air	T° wat.	O ₂ d.	pH	Cond.	Sal.	V.C.	Aqua.v.	Riv.v.	Wid.	Dep.	Chg.edg.
Station1	Mean	27.7	24.9	6.5	8.7	1105.7	0.3	56.4	0	2.0	9.5	54.7	2
	S.D.	7.5	5.7	2.0	0.5	254.0	0.2	10.3	0	0	0.7	9.2	0
Station 2	Mean	27.0	23.6	6.2	8.6	594.3	0.0	42.6	0	3.0	17.4	39.3	2
	S.D.	6.6	5.6	1.5	0.3	104.9	0.1	6.7	0	0	1.6	8.9	0
Station 3	Mean	27.4	22.6	6.8	8.5	880.5	0.2	51.0	0	2.0	14.2	46.7	3
	S.D.	6.3	3.7	1.2	0.3	153.5	0.1	17.4	0	0	0.5	1.5	0
Station 4	Mean	31.5	23.5	6.9	8.6	2351.5	1.1	51.0	0	3.0	9.5	20.7	2
	S.D.	6.4	3.5	1.2	0.4	1042.8	0.7	5.8	0	0	0.5	4.7	0
Station 5	Mean	34.0	23.6	6.4	8.7	1590.0	0.6	33.3	2.0	2.0	15.0	41.7	1
	S.D.	6.2	3.8	1.4	0.5	577.7	0.4	4.7	1.3	1.3	0	2.6	0
Station 6	Mean	31.0	23.7	7.0	8.9	1241.8	0.4	40.9	1.0	1.0	8.0	25.7	1
	S.D.	5.5	2.6	1.0	0.2	447.5	0.2	14.7	0	0	0.3	8.9	0
Station 7	Mean	29.3	22.5	4.7	8.6	1266.2	0.4	49.9	0	0	4.0	37.8	1
	S.D.	5.4	3.6	1.8	0.6	434.9	0.2	10.4	0	0	0	4.9	0
Station 8	Mean	26.0	21.1	2.6	8.3	1338.7	0.5	40.4	0.5	2.0	4.0	40.3	1
	S.D.	5.4	2.8	2.8	0.7	431.4	0.3	13.1	0.5	0	0.2	0.8	0

T° air: air temperature (°C), T° water: water temperature (°C), O₂d: Dissolved oxygen (mg/l), Cond.: Electric conductivity (μS/cm), Sal.: salinity (‰), V.C.: current velocity (cm/s), Aqua.v.: index of aquatic vegetation, Riv.v.: index of riverine vegetation, wid.: width (m), dep.: depth (cm), Chg.edg: index of changes of wadi banks. S.D.: Standard Deviation. 0: Absence; 1: Low; 2: Medium; 3: Dense/Important; Number of observations: N= 12.

In total, thirteen Odonata species were recorded. Four species were Zygopterans and nine were Anisopterans, including six Libellulids (Tab. III). No species was recorded along the watercourse during November-February.

TABLE III
Check-list of the Odonata of Wadi El Harrach

Suborder	Family	Species
Zygoptera	Calopterygidae	<i>Calopteryx haemorrhoidalis</i> (Vander Linden, 1825)
	Platycnemididae	<i>Platycnemis subdilatata</i> Selys, 1849
	Coenagrionidae	<i>Erythromma lindenii</i> (Selys, 1840)
		<i>Ischnura graellsii</i> (Rambur, 1842)
Anisoptera	Gomphidae	<i>Onychogomphus costae</i> Selys, 1885
		<i>Onychogomphus forcipatus unguiculatus</i> (Vander Linden, 1823)
	Aeshnidae	<i>Anax imperator</i> Leach, 1815
		<i>Orthetrum coerulescens anceps</i> (Schneider, 1845)
	Libellulidae	<i>Orthetrum chrysostigma</i> (Burmeister, 1839)
		<i>Sympetrum fonscolombii</i> (Selys, 1840)
		<i>Crocothemis erythraea</i> (Brullé, 1832)
		<i>Trithemis annulata</i> (Palisot de Beauvois, 1807)
		<i>Trithemis kirbyi</i> Selys, 1891

Species richness was uneven along the watercourse, varying between 2 and 9. S5 was the richest with 9 species whereas S3 hosted only 2 species. The upstream part, which had a greater vegetation cover, was noticeably poorer than the downstream part. However, this was the part where the Zygopterans were most abundant and where *Erythromma lindenii* was confined (S2). The other three species of Zygopterans were found in both upstream and downstream parts: *Calopteryx haemorrhoidalis* was observed at all stations except S7 where there was no vegetation.

The endemic but common *Platycnemis subdilatata* was observed at S1, S2 and S5. *Ischnura graellsii* was associated with stations where low vegetation was present. Likewise, three species of Anisoptera (*Onychogomphus forcipatus*, *Orthetrum coerulescens anceps* and *Orthetrum chrysostigma*), were also present both upstream and downstream.

From S4, where the habitat starts to worsen (poor water quality, low vegetation cover and almost no shading), Zygopterans were gradually replaced by Anisopterans which settled in numbers. Six Anisopteran species (*Anax imperator*, *Crocothemis erythraea*, *Onychogomphus costae*, *Sympetrum fonscolombii*, *Trithemis annulata* and *T. kirbyi*) were confined to the downstream part (from S5 downwards).

VARIATION OF THE ODONATA ASSEMBLAGES WITH ENVIRONMENTAL FACTORS

Ascending Hierarchical Classification (AHC) and the Canonical Correspondence Analysis (CCA) of the environmental and physico-chemical parameters of water suggested an association between habitat quality and the distribution of Odonata along the watercourse.

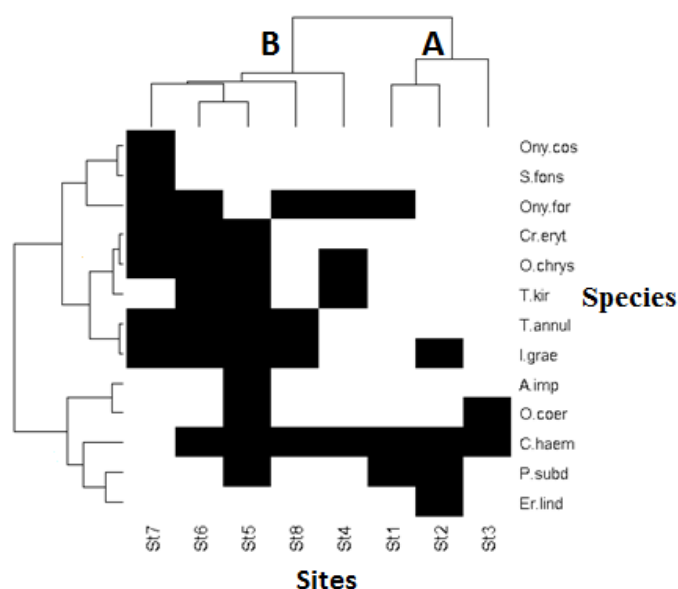


Figure 2.— Clusters based on AHC, representing the sampled stations of Wadi El Harrach: **A**: stations located upstream, and **B**: stations located downstream.

AHC indicated the presence of two distinct groups (Fig. 2). Group A included the first three upstream stations which sheltered a low number of species dominated by Zygopterans. The second group B was composed of the rest of the stations located downstream and which sheltered a higher number of species dominated by Anisopterans. Both groups may further be divided with S3 and S4 being distinct from their respective subgroups.

Results of the CCA indicated that the first two factorial axes accounted for 81.0 % of the total variation (Fig. 3). The first axis represented the altitudinal gradient with upstream stations, characterized by collinear variables like altitude, riverine vegetation, dissolved oxygen and bed width, opposed to downstream stations that harbour high values of conductivity, salinity, sunshine exposure and air temperature. The second axis spread out the upstream stations along a gradient of current velocity and conductivity/salinity. This axis separates the middle course of the wadi (S3-S6) from both the upstream part (S1 and S2) and the downstream one (S7 and S8). As was found for the AHC, the CCA organized the Odonata community along the same altitudinal gradient (first

axis) by opposing a community dominated by Zygopterans in the upstream part of the wadi to another one dominated by Anisopterans.

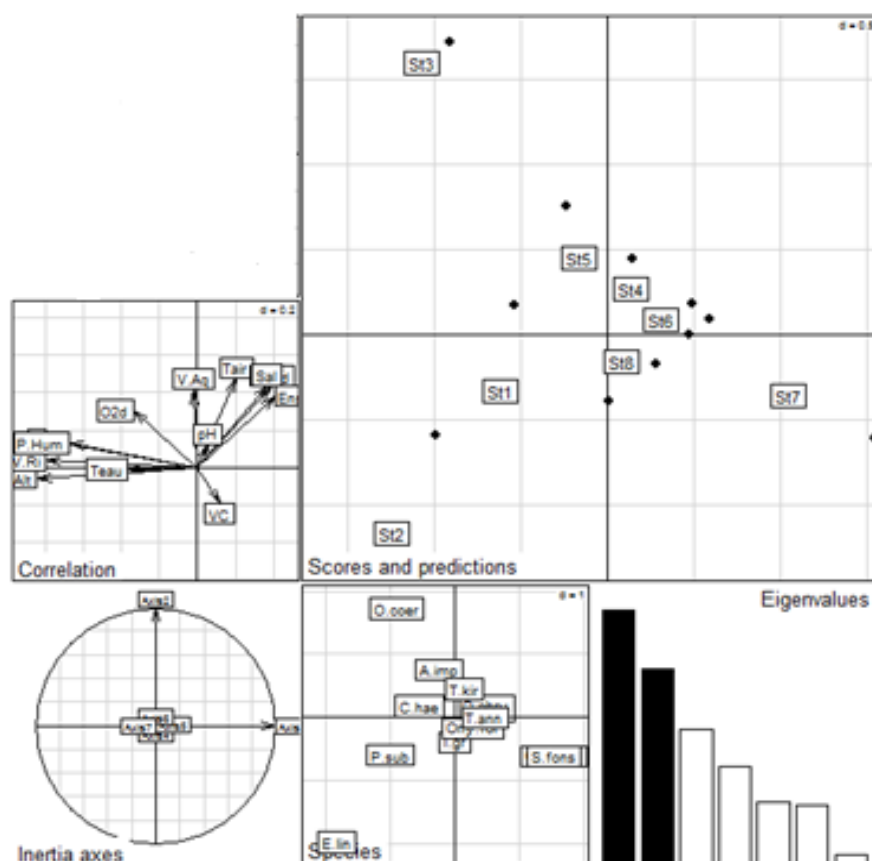


Figure 3.— Results of the Canonical Correspondence Analysis performed using recorded Odonata (presence/absence) and measured environmental parameters. The F1x2 factorial plane represents 81.0% of the total inertia.

DISCUSSION

In Algeria, like in the rest of the Maghreb, the last decades have witnessed a sizeable demographic increase (Tabutin *et al.*, 2002) but the anarchic occupation of space by both industrial and agricultural activities has had adverse consequences on the rich but vulnerable spectrum of wetlands hosted by this arid North African country (Samraoui & Samraoui, 2008; Toubal *et al.*, 2014). Thus, aquatic communities, particularly in northern Algeria, are facing up increasing pressures affecting the overall Maghrebien biodiversity and disrupting valuable ecosystem functions (Constanza, 1997; De Groot *et al.*, 2002; Wallace, 2007). Continental aquatic ecosystems, particularly watercourses, are the receptacle of urban, industrial and agricultural wastes which threaten their biota and prevent local communities from access to vital and increasingly rare freshwater. Biodiversity erosion is one of many factors that may work synergistically to compromise ecosystem services and impact human health (Chivian & Bernstein, 2008).

It comes as no surprise that the survey of the Odonata of Wadi El Harrach recovered a reduced list of species. Although sampling efforts were limited to one flight season and did not include the higher reaches of the wadi, it is possible to suggest that the overall species richness of Wadi El Harrach, amounting to 13 Odonates, is relatively poor. In other Algerian wadis like Wadi Bouarroug, that witnessed similar sampling effort, 11 species were recorded at a single station (Benchalel & Samraoui, 2012). Likewise, in a single station, 13 species were recorded at Wadi Kébir Est (Benchalel & Samraoui, 2012). A two-year survey of the watershed at Wadi Seybouse totaled 35 species (Khelifa *et al.*, 2011) whereas, with comparable sampling effort and a total of six stations, Wadi Isser provided 19 species (Bouchelouche *et al.*, 2015). However, a thorough comparison between these wadis needs to take into account their hydrology and habitat heterogeneity which may influence aquatic biodiversity (Hardersen, 2008). Sampling of exuviae and larvae would undoubtedly provide a more thorough survey of the community structures of these wadis.

Both anthropogenic and natural factors seemed to affect the Odonata assemblages of Wadi El Harrach. Physical habitat conditions like altitude (Samways, 1989; Corbet, 1999), bed width (Kietzka *et al.*, 2014), riverine vegetation (Buchwald, 1992) and shade (Remsburg *et al.*, 2008) are known to drive the distribution and abundance of Odonata riparian site selection. There is thus a need to maintain a structurally complex and heterogeneous landscape and a network of wadis exhibiting a great diversity of physical conditions (Kietzka *et al.*, 2014).

No species were recorded between November and February. Although this result agrees relatively well with the known flight period of most Mediterranean Odonates (Ferrerias-Romero & Corbet, 1995; Samraoui & Corbet, 2000), it is worth noting that a few species (*Calopteryx haemorrhoidalis*, *Lestes barbarus* and *L. numidicus*) may be seen flying in November whereas others like *Chalcolestes viridis*, *Aeshna mixta* and *Sympetrum striolatum* may extend their flight period till winter (Samraoui & Corbet, 2000; Samraoui, 2009).

The anthropogenic impact is reflected in the recreational use of the upstream parts of the wadi, the remodelling of the watercourse by channel straightening (Elosegi *et al.*, 2010), native tree logging, pine plantations (Bellot *et al.*, 2004) and invasion of alien plants like *Eucalyptus* trees (Tererai *et al.*, 2013) and Giant Reed (Boose & Holt, 1999; Quinn & Holt, 2008; Coffman *et al.*, 2010), and changes in water chemistry (Vörösmarty *et al.*, 2010) (Fig. 4).

Plant invaders and plantations have been associated with biodiversity erosion through increase of fire risks, modified hydrology, a decrease in resource levels in litter and soil processes and competition with native plant species (Richardson *et al.* 1994; Le Maitre *et al.* 2000; Brouckerhoff *et al.* 2008). It is known that land clearing for agriculture or urbanization can cause substantial increase in stream and river salinities (Williams, 1987; Kay *et al.*, 2001). Although many studies have pointed to the high tolerance of macroinvertebrates to natural salinity, there is unanimity about the lack of data on the sensitivity of freshwater invertebrates to salinity increases (Hart *et al.*, 1991). Likewise, we know little about interactions of salinity with temperature (Nelson & Palmer, 2007) or any other environmental stressors. It is known however that increased salinity may facilitate colonization of freshwater communities by invasive species (Schröder *et al.*, 2015).

All these combined anthropogenic stressors (invasive plant species, pine and *Eucalyptus* plantation, land clearing and urbanization, recreational activities, thermal pollution, increased salinity, agriculture) and their potential interactions have led to the virtual extirpation of insect life in the lowest part of Wadi El Harrach (unpublished). As was found for Wadi Isser (Bouchelouche *et al.*, 2015), the number of recorded species was highest in the middle course of the wadi. Species richness is known to peak under intermediate levels of disturbance (Connell, 1978; Wilkinson, 1999). Further down, habitat degradation was acute and, downstream of S8, the aquatic habitat was unfit for larval survival (unpublished).

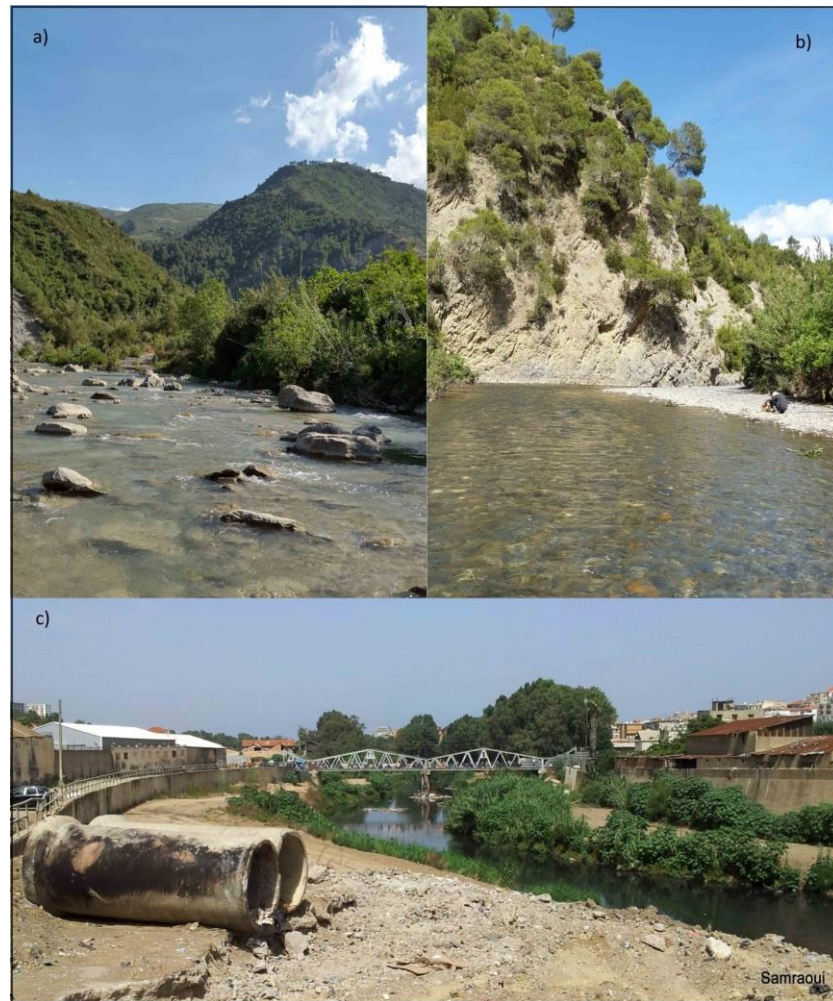


Figure 4.— A view of three stations of Wadi El Harrach: a) S1 at Lakhra and Boumaane, b) S2 at Magtaa Lazreg, c) at El Harrach (downstream of S8), a suburb of Algiers.

A further indication of the degraded state of Wadi El Harrach might be found in the IUCN North African Red List (Samraoui *et al.*, 2010): No threatened species was recorded at Wadi El Harrach during the study period. Noteworthy is the absence of two threatened endemics: *Calopteryx exul* and *Gomphus lucasi* (Riservato *et al.*, 2009). This is unsurprising as specialist species are often the most prone to extinction when their habitats are altered (Colles *et al.*, 2009). Likewise, noteworthy is the absence of other gomphids like *Onychogomphus coastae* and *Paragomphus genei*. The dominance of common species, mainly Anisopterans, was associated with the degradation of habitats in the downstream part of the wadi. Adult Anisopterans are large-bodied, accomplished insects which disperse widely (Corbet, 1999) and, in particular, recorded species like *Anax imperator*, *Crocothemis erythraea*, *Sympetrum fonscolombii*, and *Trithemis annulata* are widespread taxa, relatively tolerant of altered habitats. The presence of *Trithemis kirbyi* at Wadi El Harrach, a salt-tolerant species, previously confined to the Sahara (Samraoui & Menai, 1999), reflects the northward expansion of this dragonfly over the last decades (Boudot & Kalkman 2015).

CONCLUSION

Wadi El Harrach's poor odonatological diversity reflects the intense anthropogenic pressures it has been bearing for decades. Global changes and a concomitant loss of ecosystem services is a major source of concern as it represents a major obstacle to the Millennium objectives goals: poverty, hunger and disease reduction (Millennium Ecosystem Assessment, 2005; Patz *et al.*, 2005). In North Africa, habitat loss and degradation, water pollution, water extraction, dam construction, exotic fish introduction and human disturbance as the result of tourism and outdoor recreational activities have been identified as major threats to dragonflies and to freshwater biodiversity as a whole (Samraoui *et al.*, 2010).

Results of the physical and chemical studies are congruent with those based on the presence of Odonata and they indicate two, possibly three distinct areas: an upstream part dominated by Zygopterans associated to riverine vegetation and relatively good water quality. A downstream part dominated by salt-tolerant and ubiquitous Anisopterans, subdivided into a middle course with the highest species richness and a lower course of the wadi characterized by poor water quality and a lower species richness.

Implementing monitoring schemes and assessing regularly freshwater biodiversity are essential to resource management, thus paving the way to maintaining biological diversity and ensuring sustainable development. Such steps may include a Freshwater Action Plan for the Maghreb or frameworks like the Integrated River Basin Management (IRBM) implemented in Europe and the South African Working for Water Program. If nothing is done to avert the rapid degradation of North African freshwater ecosystems, the consequences will ultimately directly affect local people.

ACKNOWLEDGEMENTS

We are most grateful to all four referees for their helpful and instructive comments. This research was supported in part by the Algerian Ministère de l'Enseignement Supérieur et de la Recherche Scientifique and Distinguished Scientist Fellowship Program (DSFP), King Saud University, Saudi Arabia.

REFERENCES

- ABELL, R. (2002).— Conservation biology for the biodiversity crisis: a freshwater follow-up. *Conserv. Biol.*, 16: 1435-1437.
- ANNANI, F., ALFARHAN, A.H. & SAMRAOUI, B. (2012).—An ecological survey of aquatic hemiptera of northeastern Algeria (Insecta, Hemiptera). *Rev. Ecol.(Terre Vie)*, 67: 423-435.
- ARAB, A., LEK, S., LOUNACI, A. & PARK, Y.S. (2004).— Spatial and temporal patterns of benthic invertebrate communities in an intermittent river (North Africa). *Ann. Limnol. - Int. J. Lim.*, 40: 317-327.
- BELLOT, J., MAESTRE, F.T., CHIRINO, E., HERNÁNDEZ, N. & ORTIZ DE URBINA, J.M. (2004).— Afforestation with *Pinus halepensis* reduces native shrub performance in a Mediterranean semiarid area. *Acta Oecol.*, 25: 7-15.
- BENCHALEL, W. & SAMRAOUI, B. (2012).— Caractérisation écologique et biologique de l'odonatofaune de deux cours d'eau méditerranéens : l'oued El-Kébir et l'oued Bouaroug (Nord-Est de l'Algérie). *Méditerranée*, 118: 19-27.
- BETRUS, C.J., FLEISHMAN, E. & BLAIR, R.B. (2005).— Cross-taxonomic potential and spatial transferability of an umbrella species index. *J. Envir. Manage.*, 74: 79-87.
- BOOSE, A.B. & HOLT, J.S. (1999).— Environmental effects on asexual reproduction in *Arundo donax*. *Weed Res.*, 39: 117-127.
- BOUCHELOUCHE, D., KHERBOUCHE-ABROUS, O., MEBARKI, M., ARAB, A. & SAMRAOUI, B. (2015).— The Odonata of Wadi Isser (Kabylia, Algeria): Status and environmental determinants of their distribution. *Rev. Ecol. (Terre Vie)*, 70: 248-260.
- BOUDOT, J.-P. & KALKMAN, V.J. (eds) (2015).— *Atlas of the European dragonflies and damselflies of Europe*. KNNV Publishing, The Netherlands.
- BOUDOT, J.-P., KALKMAN, V.J., AZPILICUETA AMORIN, M., BOGDANOVIC, T., CORDERO RIVERA, A., DEGABRIELE, G., DOMMANGET, J.-L., FERREIRA, S., GARRIGOS, B., JOVIC, M., KOTARAC, M., LOPAU, W., MARINOV, M., MIHOKOVIC, N., RISERVATO, E., SAMRAOUI, B. & SCHNEIDER, W. (2009).— Atlas of the Odonata of the Mediterranean and North Africa. *Libellula*, Suppl. 9: 1-256.

- BROCKERHOFF, E.G., JACTEL, H., PARROTTA, J.A., QUINE, C.P. & SAYER, J. (2008).— Plantation forests and biodiversity: oxymoron or opportunity? *Biodiv. Conserv.*, 17: 925-951.
- BUCHWALD, R. (1992).— Vegetation and dragonfly fauna – Characteristics and examples of biocenological field studies. *Vegetatio*, 101: 99-107.
- CHAIB, N., BOUHALA, Z., FOUZARI, L., MARZIALI, L., SAMRAOUI, B. & ROSSARO, B., (2013a).—Environmental factors affecting the distribution of chironomid larvae of the Seybouse wadi, North-Eastern Algeria. *J. Limnol.*, 72: 203-214.
- CHAIB N., FOUZARI, A., BOUHALA Z., SAMRAOUI B. & ROSSARO B. (2013b).— Chironomid (Diptera, Chironomidae) species assemblages in northeastern Algerian hydrosystems. *J. Entomol. Acarol. Res.*, 45: 4-11.
- CHERRAIRIA, M., ADLER, P.H. & SAMRAOUI, B. (2014).— Biodiversity and bionomics of the black flies (Diptera: Simuliidae) of northeastern Algeria. *Zootaxa*, 3796: 166-174.
- CHIVIAN, E. & BERNSTEIN, A. (2008).—*How human health depends on biodiversity*. Oxford University Press, Oxford.
- CHOVANEC, A. & WARINGER, J. (2001).— Ecological integrity of river – floodplain systems assessment by dragonflies surveys (Insecta: Odonata). *Regul. River.*, 17: 493-507.
- COFFMAN, G.C., AMBROSE, R.F. & RUNDE, P.W. (2010).— Wildfire promotes dominance of invasive giant reed (*Arundo donax*) in riparian ecosystems. *Biol. Invasions*, 12: 2723-2730.
- COLLES, A., LIOW, L.H. & PRINZING, A. (2009).— Are specialists at risk under environmental change? Neoeological, paleoecological and phylogenetic approaches. *Ecol. Lett.*, 12: 849-863.
- CONNELL, J.H. (1978).— Diversity in tropical rain forests and coral reefs. *Science*, 199: 1302-1310.
- CONSTANZA, R. (1997).— The value of the world's ecosystem services and natural capital. *Nature*, 387: 253-260.
- CORBET, P.S. (1999).— *Dragonflies. Behaviour and ecology of Odonata*. Cornell University Press, Ithaca.
- DE GROOT, R.S., WILSON, M.A. & BOUMANS, R.M.J. (2002).— A typology for the classification, description and valuation of ecosystem functions, goods and services. *Ecol. Econ.*, 41: 393-408.
- DIJKSTRA, K.-D.B. & LEWINGTON, R. (2006).— *Field guide to the dragonflies of Britain and Europe*. British Wildlife Publishing, Dorset.
- DYNESIUS, M. & NILSSON, C. (1994).— Fragmentation and flow regulation of river systems in the northern third of the world. *Science*, 266: 753-762.
- ELOSEGI, A., DIEZ, J. & MUTZ, M. (2010).— Effects of hydromorphological integrity on biodiversity and functioning of river ecosystems. *Hydrobiologia*, 657: 199-215.
- FERRERAS-ROMERO, M. & CORBET, P.S. (1995).— Seasonal patterns of emergence in Odonata of a permanent stream in southwestern Europe. *Aquat. Insects*, 17: 123-127.
- FERRERAS-ROMERO, M., MÁRQUEZ-RODRIGUEZ, J. & RUIZ-GARCIA, A. (2009).— Implications of anthropogenic disturbance factors on the Odonata assemblage in a Mediterranean fluvial system. *Int. J. Odonatol.*, 12: 413-428.
- GRAND, D. & BOUDOT, J.P. (2007).— *Les libellules de France, Belgique et Luxembourg*. Biotope/Collection Parthénopé, Mèze.
- HAMZAOUI, D., HAFIANE, M., MEBARKI, M., ARAB, A., ALFARHAN, A.H. & SAMRAOUI, B. (2015).—The Gomphidae of Algeria and the Maghreb: status, ecology and conservation (Insecta: Odonata). *Int. J. Odonatol.*, 18: 175-191.
- HARDERSEN, S. (2008).— Dragonfly (Odonata) communities at three lotic sites with different hydrological characteristics. *Ital. J. Zool.*, 75: 271-283.
- HART, B.T., BAILEY, P., EDWARDS, R., HORTLE, K., JAMES, K., MCMAHON, A., MEREDITH, C. & SWADLING, K. (1991).— A review of the salt sensitivity of the Australian freshwater biota. *Hydrobiologia*, 210: 105-144.
- KAY, W.R., HALSE, S.A., SCANLON, M.D. & SMITH, M.J. (2001).— Distribution and environmental tolerances of aquatic macroinvertebrate families in the agricultural zone of southwestern Australia. *J. N. Am. Benthol. Soc.*, 20: 182-199.
- KHELIFA, R., YUCEFI, A., KAHLERRAS, A., ALFARHAN, A.H., AL-RASHEID, K.A.S. & SAMRAOUI, B. (2011).— L'odonatofaune de la Seybouse en Algérie: intérêt pour la biodiversité du Maghreb (Insecta: Odonata). *Rev. Ecol. (Terre Vie)*, 66: 55-66.
- KIETZKA, G.J., PRYKE, J.S. & SAMWAYS, M.J. (2014).— Landscape ecological networks are successful in supporting a diverse dragonfly assemblage. *Insect Conserv. Div.*, 8: 229-237.
- LE MAITRE, D.C., VERSFELD, D.B. & CHAPMAN, R.A. (2000).— The impact of invading alien plants on surface water resources in South Africa: a preliminary assessment. *Water SA*, 26: 397-408.
- LOUNACI, A., BROSE, S., MOULOUD, S.A., LOUNACI-DAOUDI, D., MEBARKI, M. & THOMAS, A. (2000a).— Current knowledge of benthic invertebrate diversity in Algerian stream: a species check-list of the Sébaou river basin (Tizi Ouzou). *Bull. Soc. Hist. Nat. Toulouse*, 136: 43-55.
- LOUNACI, A., BROSE, S., THOMAS, A. & LEK, S. (2000b).— Abundance, diversity and community structure of macroinvertebrates in an Algerian stream: the Sébaou wadi. *Ann. Limnol.*, 36: 123-133.
- MILLENNIUM ECOSYSTEM ASSESSMENT (2005).— *Ecosystems and Human Well-being: General Synthesis*. Island Press and World Resources Institute, Washington, DC.

- MONTEIRO-JÚNIOR, C.S., JÜEN, L. & HAMADA, N. (2015).— Analysis of urban impacts on aquatic habitats in the central Amazon basin: Adult odonata as bioindicators of environmental quality. *Ecol. Indic.*, 48: 303-311.
- NELSON, K.C. & PALMER, M.A. (2007).— Stream temperature surges under urbanization and climate change: data, models, and responses. *J. Am. Water Resour. Assoc. (JAWRA)*, 43: 440-452.
- OERTLI, B. (2008).— The use of dragonflies in the assessment and monitoring of aquatic habitats. Pp 79-95 In: A. Córdoba-Aguilar (ed.). *Dragonflies: Model organisms for ecological and evolutionary research*. Oxford University Press, Oxford.
- OLSON, D.M. & DINERSTEIN, E. (1998).— The global 2000: a representation approach to conserving the earth's most biological valuable ecoregions. *Conserv. Biol.*, 12: 502-515.
- PATZ, J.A., CAMPBELL-LENDRUM, D., HOLLOWAY, T. & FOLEY, J.A. (2005).— Impact of regional climate change on human health. *Nature*, 438: 310-317.
- QUINN, L.D. & HOLT, J.S. (2008).— Ecological correlates of invasion by *Arundo donax* in three southern California riparian habitats. *Biol. Invasions*, 10: 591-601.
- R DEVELOPMENT CORE TEAM (2014).— *R: A language and environment for statistical computing*. Vienna, Austria.
- REMSBURG, A.J., OLSON, A.C. & SAMWAYS, M.J. (2008).— Shade alone reduces adult dragonfly (Odonata: Libellulidae) abundance. *J. Insect. Behav.*, 21: 460-468.
- RICHARDSON, D.M., WILLIAMS, P.A. & HOBBS, R.J. (1994).— Pine invasions in the southern Hemisphere: determinants of spread and invadability. *J. Biogeogr.*, 21: 511-527.
- RISERVATO, E., BOUDOT, J.-P., FERREIRA, S., JOVIĆ, M., KALKMAN, V.J., SCHNEIDER, W., SAMRAOUI, B. & CUTTELOD, A. (2009).— *The status and distribution of dragonflies of the Mediterranean Basin*. IUCN, Gland.
- SAMRAOUI, B. (2009).— Seasonal ecology of Algerian Lestidae (Odonata). *Int. J. Odonatol.*, 12: 383-394.
- SAMRAOUI, B., BOUDOT, J.-P., FERREIRA, S., RISERVATO, E., JOVIĆ, M., KALKMAN, V.J. & SCHNEIDER, W. (2010).— The status and distribution of dragonflies. Pp. 51-70 In: N. Garcia, A. Cuttelod & D. Abdul Malak (eds). *The status and distribution of freshwater biodiversity in Northern Africa*. IUCN, Gland, Switzerland, Cambridge, UK, and Malaga, Spain.
- SAMRAOUI, B. & CORBET, P.S. (2000).— The Odonata of Numidia, Northeastern Algeria, Part II Seasonal Ecology. *Int. J. Odonatol.*, 3: 27-39.
- SAMRAOUI, B. & MENAI, R. (1999).— A contribution to the study of Algerian Odonata. *Int. J. Odonatol.*, 2: 145-165.
- SAMRAOUI, B. & SAMRAOUI, F. (2008).— An ornithological survey of the wetlands of Algeria: Important Bird Areas, Ramsar sites and threatened species. *Wildfowl*, 58: 71-98.
- SAMWAYS, M.J. (1989).— Taxon turnover in Odonata across a 3000 m altitudinal gradient in southern Africa. *Odonatologica*, 18: 263-274.
- SCHMINKE, H.K. (2007).— Entomology for the copepodologist. *J. Plankton Res.*, 29: 149-162.
- SCHRÖDER, M., SONDERMANN, M., SURES, B. & HERING, D. (2015).— Effects of salinity gradients on benthic invertebrate and diatom communities in a German lowland river. *Ecol. Indic.*, 57: 236-248.
- SIMAIKA, J.P. & SAMWAYS, M.J. (2008).— Valuing dragonflies as service providers. Pp 109-123 In: A. Córdoba-Aguilar (ed.). *Dragonflies: Model organisms for ecological and evolutionary research*. Oxford University Press, Oxford.
- SIMAIKA, J.P. & SAMWAYS, M.J. (2009).— An easy-to-use index of ecological integrity for prioritizing freshwater sites and for assessing habitat quality. *Biodiv. Conserv.*, 18: 1171-1185.
- TABUTIN, D., VILQUIN, E. & BIRABEN, J.-N. (2002).— *L'histoire de la population de l'Afrique du Nord pendant le deuxième millénaire*. Université catholique de Louvain, Département des sciences de la population et du développement, Document de travail, 15. Louvain-la-Neuve.
- TERERAI, F., GAERTNER, M., JACOBS, S.M. & RICHARDSON, D.M. (2013).— *Eucalyptus* invasions in riparian forests: effects on native vegetation community diversity, stand structure and composition. *Forest Ecol. Manage.*, 297: 84-93.
- THOMAS, A. & LOUNACI, A. (1989).— Compléments et corrections à la faune des Ephéméroptères d'Afrique du Nord. 4. Les stades ailés de *Baetis punicus* Thomas, Boumaiza et Soldan, 1983 (Baetidae). *Bull. Soc. Hist. Nat. Toulouse*, 125: 27-29.
- TOUBAL, O., BOUSSEHABA, A., TOUBAL, A. & SAMRAOUI, B. (2014).— Biodiversité méditerranéenne et changements globaux: cas du complexe de zones humides de Guerbes-Senhadja, Algérie. *Physio-Géo*, 8: 273-295.
- VÖRÖSMARTY, C.J., MCINTYRE, P.B., GESSNER, M.O., DUDGEON, D., PRUSEVITCH, A., GREEN, P., GLIDDEN, S., BUNN, S.E., SULLIVAN, C.A., REIDY LIERMANN, C. & DAVIES, P.M. (2010).— Global threats to human water security and river biodiversity. *Nature*, 467: 555-561.
- WILLIGALLA, C. & FARTMANN, T. (2012).— Patterns in the diversity of dragonflies (Odonata) in cities across Central Europe. *Eur. J. Entomol.*, 109: 235-245.
- WALLACE, K.J. (2007).— Classification of ecosystem services: problems and solutions. *Biol. Conserv.* 39: 235-246.
- WILKINSON, D.M. (1999).— The disturbing history of intermediate disturbance. *Oikos*, 84: 145-147.
- WILLIAMS, W.D. (1987).— Salinization of rivers and streams: an important environmental hazard. *Ambio*, 16: 180-185.